Lecture 10b: Forecasting the Power Grid

Outline

- Background
- Project goals
- Getting started

Background

Load Forecasting

- The amount of electricity drawn from the power grid (the "load") varies throughout the day, and from day to day
- Companies that draw a lot of electrical power from the grid are interested in forecasting the load
- Why?

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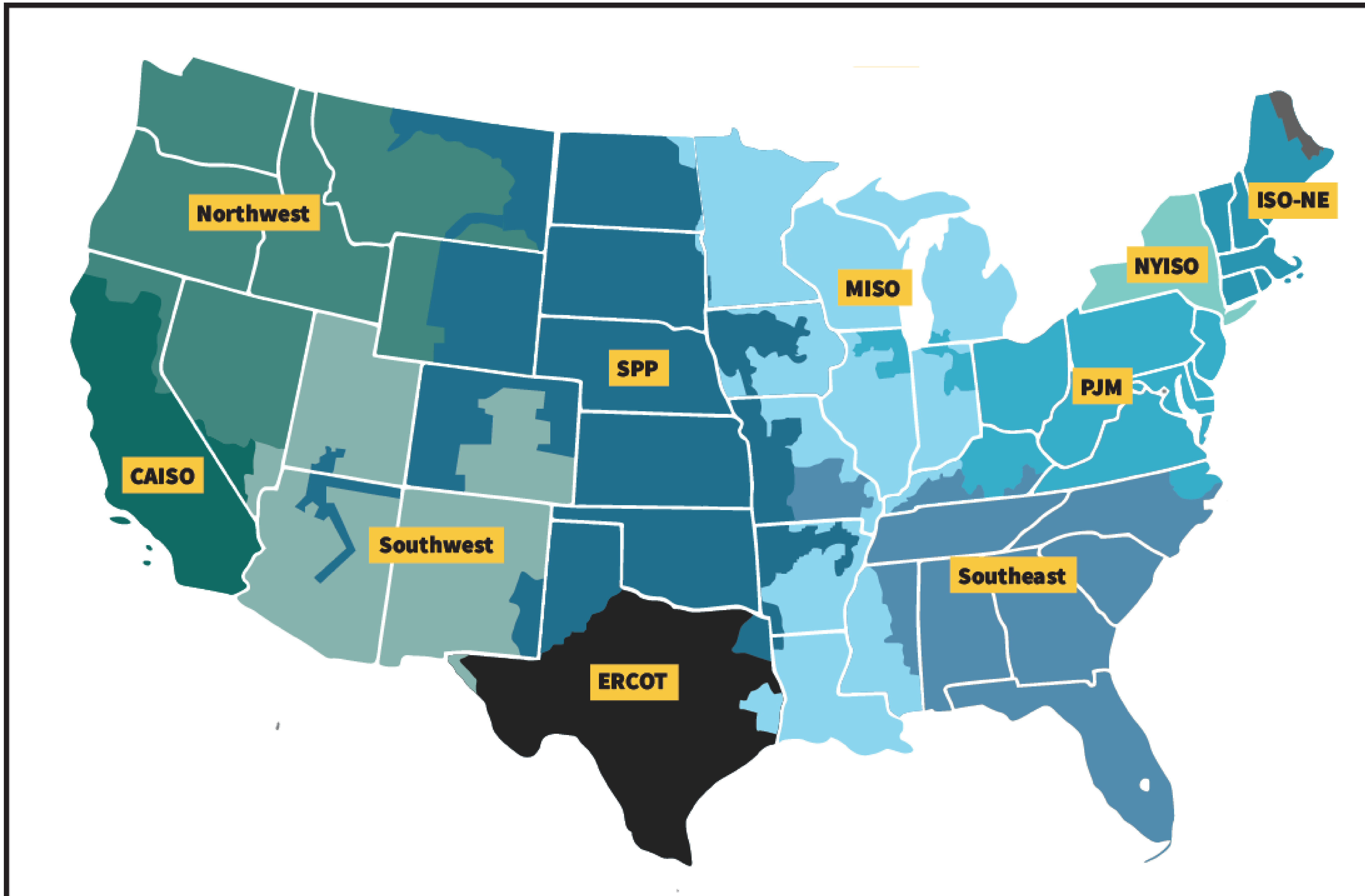
- Pricing!

Industrial power pricing

- Industrial power customers (eg, refrigeration) are typically charged in multiple ways:
    - Overall power usage (KW hr)
    - Power usage during each day's hour of peak load
    - Power usage during the peak hour(s) of load of the year

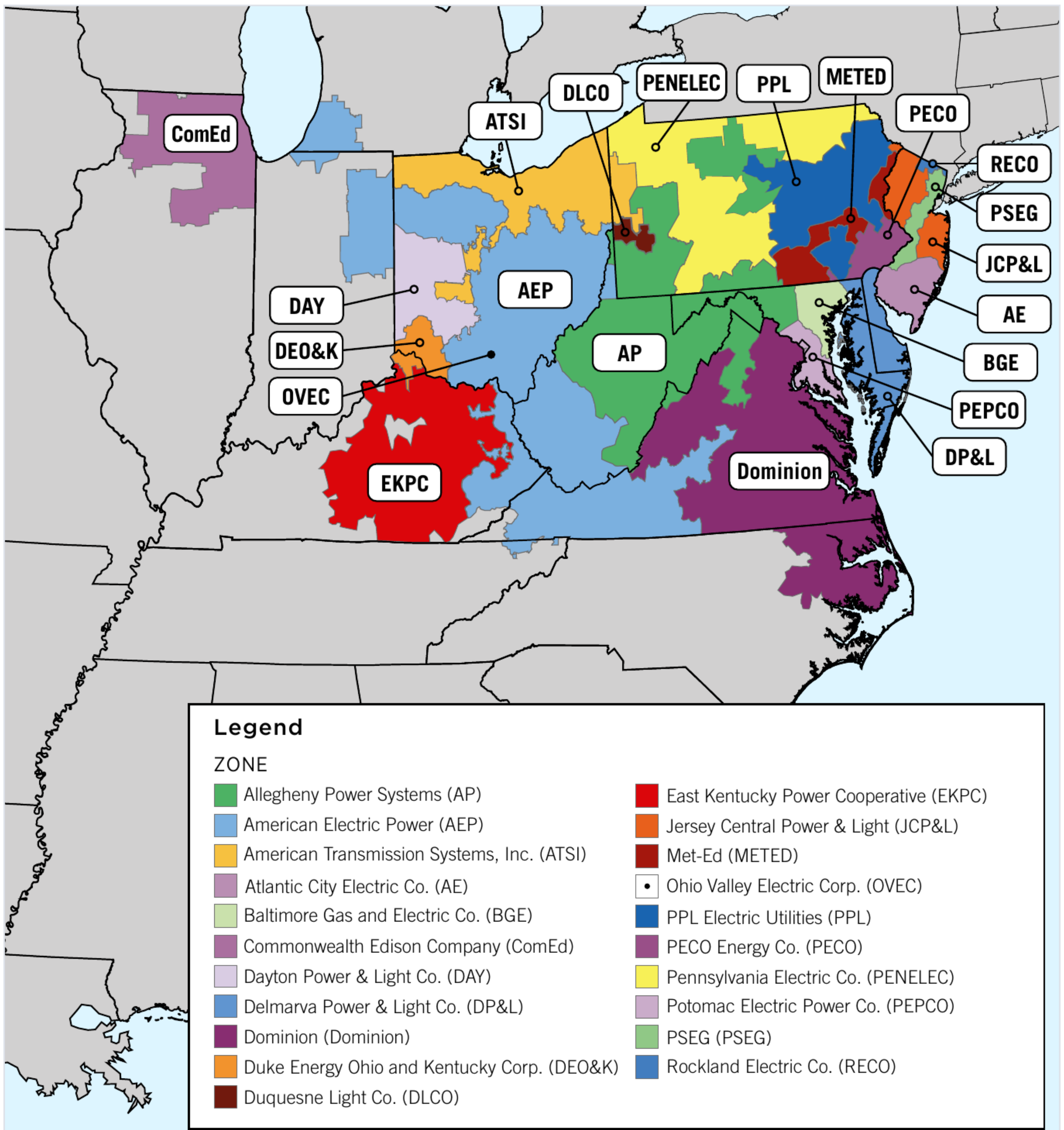
The US power grid

The grid is divided into several regional transmission organizations (RTOs):



PJM

We will focus on PJM ("Pennsylvania-New Jersey-Maryland"), which is itself divided into several zones:



29 zones:

Zones
AECO
AEPAPT
AEPIMP
AEPKPT
AEPOPT
AP
BC
CE
DAY
DEOK
DOM
DPLCO
DUQ
EASTON
EKPC
JC
ME
OE
OVEC
PAPWR
PE

Data

PJM portal:

https://dataminer2.pjm.com

Historical "metered" data:

https://dataminer2.pjm.com/feed/hrl_load_metered

Project goals

Overview

Three tasks:

- Forecast hourly loads
- Predict which hour of the day will have the peak load (the "peak hour")
- Predict which days will have maximum peak loads (the "peak days")

Prediction schedule

- Predictions will be made each day beginning Wednesday, November 19 and ending Friday, November 28 (inclusive)
- Predictions will be for the following day
- Predictions are due at noon
- Your first predictions are due at noon on Nov 19, and will be for Nov 20
- Your final predictions are due at noon on Nov 28 and will be for Nov 29

Task 1: Forecast hourly loads

- Forecast the hourly load (in MW) in each of 29 regions
- Goal is to minimize mean squared error
- You will make a total of 6,960 predictions: (29 regions) x (10 days) x (24 hours) = 6,960

Predict which hour of the day will have the peak load

- For each day and each region, define the "peak hour" as the hour with the maximum load (for that day and region)
- For each day and each region, you will predict the "peak hour"
- Simple approach: pick the hour with the maximum forecasted load
- "Success": If your prediction is correct, +/- one hour; simple approach not necessarily optimal
- You will make a total of 290 predictions: (29 regions) x (10 days) = 290

Predict which days will have maximum peak loads

- For each region, define the "peak days" as the 2 days out of the 10 with the highest load during the peak hour
- Note that peak days are not actually known until the end of the 10 days
- For each day and each region, predict whether the day will be a peak day
- Loss function:

	Predicted peak day	**Predicted non-peak day**
Actual peak day	0	4
Actual non-peak day	1	0

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- Each day you will make 754 predictions: 29 x 24 (hourly load) + 29 (peak hour) + 29 (peak day)

Project timeline

- Tuesday, Nov 18: Model finalized, code committed to Github, Docker image uploaded to Dockerhub
- Wednesday, Nov 19: Begin making daily predictions, due at noon daily
- Thursday, Nov 20: Speed presentations
- Friday, Nov 28: Final day making predictions
- TBD: Longer presentations, by invitation

Data and analysis

- Any publicly available data you want
- Any methods you want

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- Load prediction models

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- Load prediction models

Discuss: What data might you want?

Reproducibility

- Compose all analyses in a markdown notebook environment
- Organize code into a coherent workflow
    - Use makefiles
    - Cache intermediate outputs
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- Save entire analysis in docker image
- Docker image can also be run to make live predictions

Docker image

- Docker image should be uploaded to Dockerhub, and shared with me
- Should include:
    - Raw data (exactly as downloaded from the internet)
    - All code, including to:
        - Download raw data
        - Fit models
        - Download current conditions and make predictions
    - Intermediate outputs, e.g.,
        - Cleaned/processed data
        - Fitted models
- Should support `linux/amd64` platform
- Recommended base image: `jupyter/r-notebook`

Running the Docker image

- The command

  `docker run -it --rm yourdockerhubname/yourimagename`

  should provide a bash terminal where commands can be run to reproduce the analysis (see next slide)
- The command

  `docker run -it --rm yourdockerhubname/yourimagename make predictions`

  should output that day's predictions to the screen, and then quit

Running the Docker image (default terminal)

- The command

  `docker run -it --rm yourdockerhubname/yourimagename`

  should provide a bash terminal where commands can be run to reproduce the analysis
- The following commands should be available:
  - `make clean` -- deletes everything except for the code (i.e., markdown files) and raw data (as originally downloaded)
  - `make` -- runs all analyses (except downloading raw data and making current predictions)
  - `make predictions` -- makes current predictions and outputs them to the screen
  - `make rawdata` -- deletes and re-downloads the raw data
- Other make commands, if/as appropriate

Running the Docker image (predictions)

- The command
  `docker run -it --rm yourdockerhubname/yourimagename make predictions`
  should output that day's predictions to the screen, and then quit
- The output should look like:
  `"YYYY-MM-DD", L1_00, L1_01, L1_02, ..., L1_23, L2_00, ..., L2_23, ..., L_29_23, PH_1, PH_2, ..., PH_29, PD_1, ..., PD_29` where
  - `"YYYY-MM-DD"` is the current date
  - `L1_00` is the predicted load of zone 1 form 00:00 to 01:00 (midnight to 1am), in megawatts
    - Should be rounded to the nearest integer
  - `PH_1` is the predicted peak hour of zone 1 (00 to 23)
  - `PD_1` is the prediction of whether zone 1 will have a peak day (0 for no, 1 for yes)
- **No other output should be displayed**

Recording your predictions

Every day before noon you should:

- Run `docker run -it --rm yourdockerhubname/yourimagename make predictions >> predictions.csv`
- Commit the updated `predictions.csv` to github
- Add your new predictions to the class's Google Sheets document (OK if completed a little after noon)

Computational resources

- Your docker image should be no larger than 5GB
- `make` should run in under 4 hours, assuming
  - 4 cores
  - 32GB RAM
  - No GPU
- `make predictions` should run in under 5 minutes, assuming
  - 1 core
  - 4GB RAM
  - MWireless connection

Getting started

First steps?

Discuss

First steps?

Some recommended first steps:

1. Create a docker that outputs *something* in the required format (even all 0s!) (In the next few days)
2. Create a base model that outputs the historical average of each region / hour (By end of next week)
3. Improve
    - day of week
    - time of year
    - recent load (say, previous day or week)
    - weather

Collaboration

- Collaboration is allowed, but **no cliques**
- All discussion should be public on Canvas
- Code must be shared via github
  - github must be public, posted on Canvas
  - borrowed code must be documented as such

Gen-AI

- Allowed
- Must be clearly where used
- Analysis must still be well documented and organized